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Dual-Frequency Power Distribution Method for Omnidirectional Wireless Power Transfer System

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Keywords: dual-frequency | omnidirectional wireless power transfer (OWPT) | power distribution | wireless power transmitter

ABSTRACT

Omnidirectional wireless power transfer (OWPT) systems are widely used in consumer electronic products (CEPs), industrial robots, medical devices, and more. However, the different CEPs usually have different power and frequency requirements to ensure various transmission characteristics. In order to meet the transmission requirements of CEPs with different power and frequency needs, this paper proposes a novel dual-frequency power allocation method for OWPT. This method achieves independent regulation of output power for loads of different frequencies through the designed SS topology dual-frequency resonant compensation structure and establishes corresponding parameter design methods. A dual-frequency OWPT platform operating at 40 kHz/100 kHz was built to verify the correctness and effectiveness of the power allocation method. Experimental results show that when the input voltage corresponding to only one frequency changes, only the output power of the load corresponding to that frequency changes accordingly. Meanwhile, under the condition of dual-load power supply, when the receiving coil is within a 90° offset range, the system's output power ranges from 49.25 to 57.15 W with a fluctuation rate of 14.8%, and the output efficiency ranges from 76.67% to 78.68% with a fluctuation rate of 2.59%.

1 | Introduction

Due to its convenience and intelligence, the consumer electronic products (CEPs) have become one of the most widely used electronic devices today. Smartphones, smartwatches, and wireless earphones all fall into the category of CEP. As these CEPs continue to become more intelligent and portable, the issue of battery capacity is becoming increasingly severe. Omnidirectional wireless power transfer (OWPT) systems provide a highly convenient and flexible new method of power supply for CEP by enhancing the spatial uniformity of the magnetic field or strengthening the intensity of the directional magnetic field [1–3]. Currently, research on OWPT systems for CEP mainly focuses on inductive coupling-based wireless energy transfer [4, 5]. Reasonable power allocation and regulation can improve the overall performance of omnidirectional WPT systems, making them more adaptable and flexible to meet various environmental conditions and application needs [6–9]. Therefore, flexibly meeting the power and frequency requirements of different load devices is a key issue in the current design of OWPT systems [10–12].

In the study of OWPT systems, many institutions have deeply explored the optimization design of omnidirectional coupler

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structures and methods for power distribution and regulation. The team led by Chun T. Rim proposed various two-dimensional planar coil structures capable of creating a large area of uniform magnetic fields to power multiple electrical devices [13–15]. Hui introduced a triorthogonal circular coil structure [16] and a dual-layer triorthogonal square coil transmitting structure [17], which respectively achieve omnidirectional and uniform magnetic fields, providing high-degree dynamic wireless charging for space drones. By combining topology circuits with orthogonal coils, a stable rotating magnetic field is generated. Feng and others proposed a bowl-shaped structure composed of multiple transmitting coils to meet the power needs of small electronic devices [18]. The team led by Wang proposed a cylindrical TC structure powered by three independent groups, achieving stable output power [19]. Wang's team studied the adaptive coupling on the receiving side of OWPT, achieving wide-range adaptive reception on the receiver side [20, 21]. Despite the achievements in electrical energy reception for devices within a certain range, issues such as poor energy concentration, low overall system efficiency, and severe magnetic leakage in nonpowered areas remain. These problems can significantly interfere with the reliable power supply and system stability of non-powered area devices.

To achieve power regulation for multiple loads in space, Dang designed a triorthogonal ring helical coil transmitting structure [22]. This design adjusts the power in each channel based on the coupling between each channel and the receiving coil, enabling energy transmission to any load within the spatial range. However, this design cannot simultaneously regulate the power for multiple receiving loads. Ma proposed a scalable planar array transmitting coil structure that can charge any receiving coil in six degrees of freedom within the space [23]. To meet the energy transmission needs of more loads, additional transmitting coils need to be added to expand the charging coverage. However, this system cannot achieve independent regulation for multiple loads. Song utilized a time-division multiplexing method to realize an OWPT system for charging multiple devices with different power requirements [24]. However, the three-coil structure limits the number of devices that can be charged simultaneously, and the charging order of the loads depends on their priority. Liu proposed a dual-frequency OWPT system for smart home sensors. This system consists of a dual-frequency three-dimensional transmitting coil structure and two planar receiving coils, capable of transmitting power to two receiving coils of different frequencies simultaneously in space [25]. However, the use of a relay coil structure in this system leads to complex coupling relationships, increasing the design requirements for system parameters. Zhu proposed a dual-frequency OWPT system based on triorthogonal coils to simultaneously drive two independent loads and achieve independent voltage regulation [26]. However, this system requires three sets of inverters and compensation networks, which increases the system cost, losses, and control complexity.

In summary, in order to achieve simultaneous power supply to multiple CEP loads within a space and to flexibly adjust power and frequency to meet the needs of different CEP devices, reducing system cost, loss, and control complexity, while maximizing the utilization of the spatial magnetic field to minimize magnetic leakage loss, is the current focus of research on power allocation and regulation methods for OWPT systems.

To address the aforementioned issues, this paper proposes a dual-frequency power distribution method and corresponding parameter design method for OWPT systems. An omnidirectional dual-frequency power distribution system is established based on inner right-angle region omnidirectional transmitting coils, achieving selective reception of transmitted power and independent regulation of output power.

2 | System Characteristic Analysis and Topology Parameter Design

2.1 | Circuit Analysis

The schematic diagram of the OWPT system based on dual-frequency resonance is shown in Figure 1. This paper takes the SDT structure and its optimal efficiency strategy reported in [27] as an example. This structure consists of three identical square coils and a ferrite shielding layer. It is decoupled and optimized through a series-connected bipolar coupling coil to enhance the SDT structure. This effectively increases the coupling strength of the system, improves system efficiency, and significantly reduces system leakage in non-working areas.

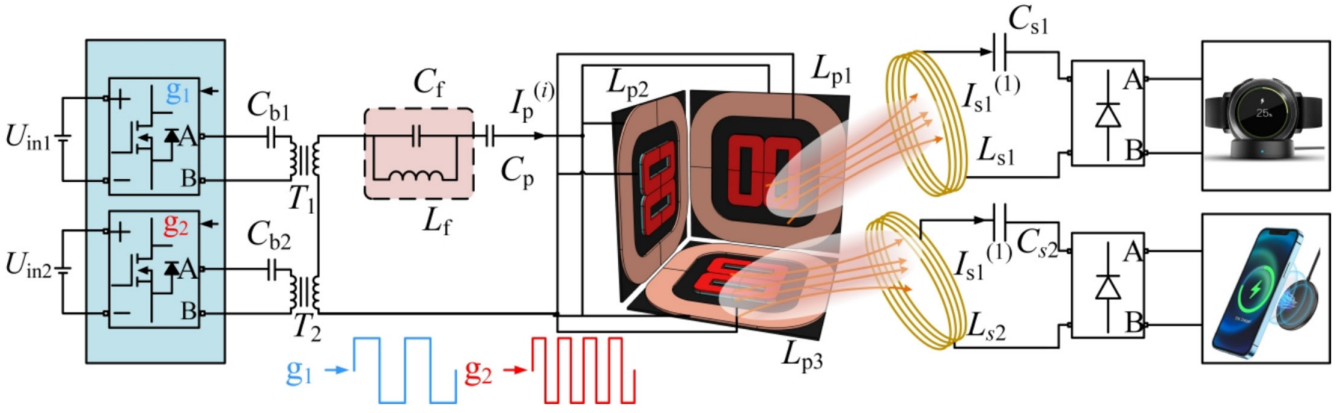


FIGURE 1 | Schematic of the dual-frequency resonance OWPT system.

Two DC power supplies are connected to two inverters with different switching frequencies to drive the system. The two inverters respectively output square wave voltages with frequencies of f_1 and f_2 and achieve superposition on the transmitter coil (TX) side through their respective isolation transformers T_1 and T_2, thus realizing dual-frequency composite excitation of the TX coil. The transmitting side of the system consists of a dual-frequency resonance network and a TX coil, while the receiving side contains two receiving circuits, each of which is composed of a receiving coil (RX), a compensation capacitor, and a load.

The equivalent mutual inductance model of the dual-frequency omnidirectional WPT system is shown in Figure 2. $\boldsymbol{U}_{\text{in}}^{(1)}$ and $\boldsymbol{U}_{\text{in}}^{(2)}$ are two AC square wave input voltages of different frequencies that are superimposed through isolation transformers, operating at frequencies f_1 and f_2, respectively. The TX is composed of three square subcoils connected in parallel, with the three subcoil units orthogonally adjacent to each other, as shown in Figure 1. The self-inductance of the TX is L_{p}. The series dual-frequency resonance compensation network consists of C_{p}, C_{f}, and L_{f}, which can achieve resonance at both frequencies simultaneously, enabling MULTICHANNEL transmission in the OWPT system. L_{s1} and L_{s2} are the self-inductances of the two receiving coils RX_{s1} and RX_{s2} respectively. Both receiving circuits use a series compensation topology, with compensation capacitors C_{s1} and C_{s2}, respectively. L_{s1} and C_{s1} form a frequency f_1 selective network, while L_{s2} and C_{s2} form a frequency f_2 selective network. $\boldsymbol{I}_{\text{p}}$, $\boldsymbol{I}_{\text{s1}}$, and $\boldsymbol{I}_{\text{s2}}$ are the currents flowing through the TX, RX_{s1}, and RX_{s2} respectively. M_{p1s} and M_{p2s} are the mutual inductances between the TX and the RX_{s1} and RX_{s2}, respectively. R_{p}, R_{s1}, and R_{s2} are the equivalent internal resistances of the TX, RX_{s1}, and RX_{s2}, respectively. R_{eq1} and R_{eq2} are the equivalent resistances of the two receiving circuits viewed from the rectifier input side.

According to the superposition principle, the load reception power can be considered as the sum of the load reception power when each of the two power sources acts independently. Using the fundamental wave analysis method to analyze the dual-frequency system, the decoupled equivalent circuit of the system is shown in Figure 3. $\boldsymbol{I}_{\text{p}}^{(2)}$, $\boldsymbol{I}_{\text{s1}}^{(2)}$, and $\boldsymbol{I}_{\text{s2}}^{(2)}$ are the currents flowing through the TX, RX_{s1}, and RX_{s2} respectively when $\boldsymbol{U}_{\text{in}}^{(1)}$ acts independently, with an operating angular frequency of ω_1; $\boldsymbol{I}_{\text{p}}^{(2)}$, $\boldsymbol{I}_{\text{s1}}^{(2)}$, and $\boldsymbol{I}_{\text{s2}}^{(2)}$ are the currents flowing through the TX, RX_{s1}, and RX_{s2}, respectively, when $\boldsymbol{U}_{\text{in}}^{(2)}$ acts independently, with an operating angular frequency of ω_2. According to KVL, the current loop equations for the circuit shown in Figure 2 are

$$\begin{bmatrix} Z_{\text{p}}^{(i)} & -\text{j}\omega_i M_{\text{ps1}} & -\text{j}\omega_i M_{\text{ps2}} \\ -\text{j}\omega_i M_{\text{ps1}} & Z_{\text{s1}}^{(i)} & \text{j}\omega_i M_{\text{s1s2}} \\ -\text{j}\omega_i M_{\text{ps2}} & \text{j}\omega_i M_{\text{s1s2}} & Z_{\text{s2}}^{(i)} \end{bmatrix} \begin{bmatrix} \boldsymbol{I}_{\text{p}}^{(i)} \\ \boldsymbol{I}_{\text{s1}}^{(i)} \\ \boldsymbol{I}_{\text{s2}}^{(i)} \end{bmatrix} = \begin{bmatrix} \boldsymbol{U}_{\text{in}}^{(i)} \\ -R_{\text{eq1}}\boldsymbol{I}_{\text{s1}}^{(i)} \\ -R_{\text{eq2}}\boldsymbol{I}_{\text{s2}}^{(i)} \end{bmatrix} \tag{1}$$

where $i = 1, 2$, and the impedances $Z_{\text{p}}^{(i)}$, $Z_{\text{s1}}^{(i)}$, and $Z_{\text{s}}^{(i)}$ are

$$\begin{cases} Z_{\text{p}}^{(i)} = \text{j}\omega_i L_{\text{p}} + \dfrac{1}{\text{j}\omega_i C_{\text{p}}} + \dfrac{\text{j}\omega_i L_{\text{f}}}{1-\omega_i^2 L_{\text{f}} C_{\text{f}}} \\ Z_{\text{s1}}^{(i)} = \text{j}\omega_i L_{\text{s1}} + \dfrac{1}{\text{j}\omega_i C_{\text{s1}}} \\ Z_{\text{s2}}^{(i)} = \text{j}\omega_i L_{\text{s2}} + \dfrac{1}{\text{j}\omega_i C_{\text{s2}}} \end{cases} \tag{2}$$

From Equations (1) and (2), the currents for each coil can be obtained as follows:

$$\begin{cases} \boldsymbol{I}_{\text{p}}^{(i)} = \dfrac{\boldsymbol{U}_{\text{in}}^{(i)}\sigma}{\delta} \\ \boldsymbol{I}_{\text{s1}}^{(i)} = \dfrac{\boldsymbol{U}_{\text{in}}^{(i)}\omega_i\sigma}{\delta} \\ \boldsymbol{I}_{\text{s2}}^{(i)} = \dfrac{\boldsymbol{U}_{\text{in}}^{(i)}\omega_i\sigma}{\delta} \end{cases} \tag{3}$$

where

$$\sigma = \left(M_{\text{s1s2}}^2\omega^2 + \left(R_{\text{eq1}} + Z_{\text{s1}}^{(i)}\right)\left(R_{\text{eq2}} + Z_{\text{s2}}^{(i)}\right)\right) \tag{4a}$$

$$\begin{aligned} \delta = &-2\text{j}M_{\text{ps1}}M_{\text{ps2}}M_{\text{s1s2}}\omega_i^3 + M_{\text{ps2}}^2\omega_i^2\left(R_{\text{eq1}} + Z_{\text{s1}}^{(i)}\right) \\ &+ M_{\text{ps1}}^2\omega^2\left(R_{\text{eq2}} + Z_{\text{s2}}^{(i)}\right) + Z_{\text{p}}^{(i)}\sigma \end{aligned} \tag{4b}$$

According to Figure 3, using the superposition theorem, the output power of the system under dual-frequency composite excitation, P_{o1} and P_{o2}, can be obtained as follows:

$$\begin{cases} P_{\text{o1}} = \displaystyle\sum_{i=1}^{2}\left|I_{\text{s1}}^{(i)}\right|^2 R_{\text{eq1}} \\ P_{\text{o2}} = \displaystyle\sum_{i=1}^{2}\left|I_{\text{s2}}^{(i)}\right|^2 R_{\text{eq2}} \end{cases} \tag{5}$$

where $P_{\text{o1}}^{(1)}$ and $P_{\text{o1}}^{(2)}$ represent the output power components of receiving circuit 1 under the influence of frequencies f_1 and f_2, respectively, and P_{o1} represents the total output power of receiving circuit 1; $P_{\text{o2}}^{(1)}$ and $P_{\text{o2}}^{(2)}$ represent the output power components of receiving circuit 2 under the influence of frequencies f_1 and f_2, respectively, and P_{o2} represents the total output power of receiving circuit 2.

The power loss of the system's transmitting and receiving circuits is mainly due to the internal resistance loss of the coupling coils. Assuming the internal resistances of the TX, RX_{s1}, and

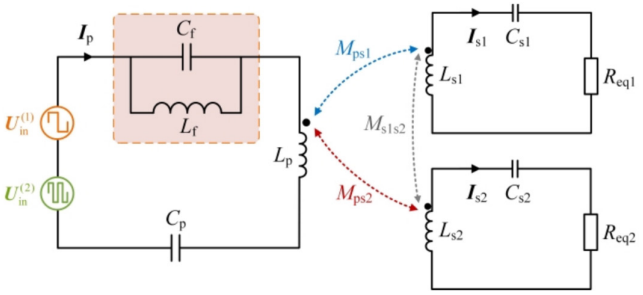


FIGURE 2 | Equivalent circuit of the dual-frequency resonance OWPT system.

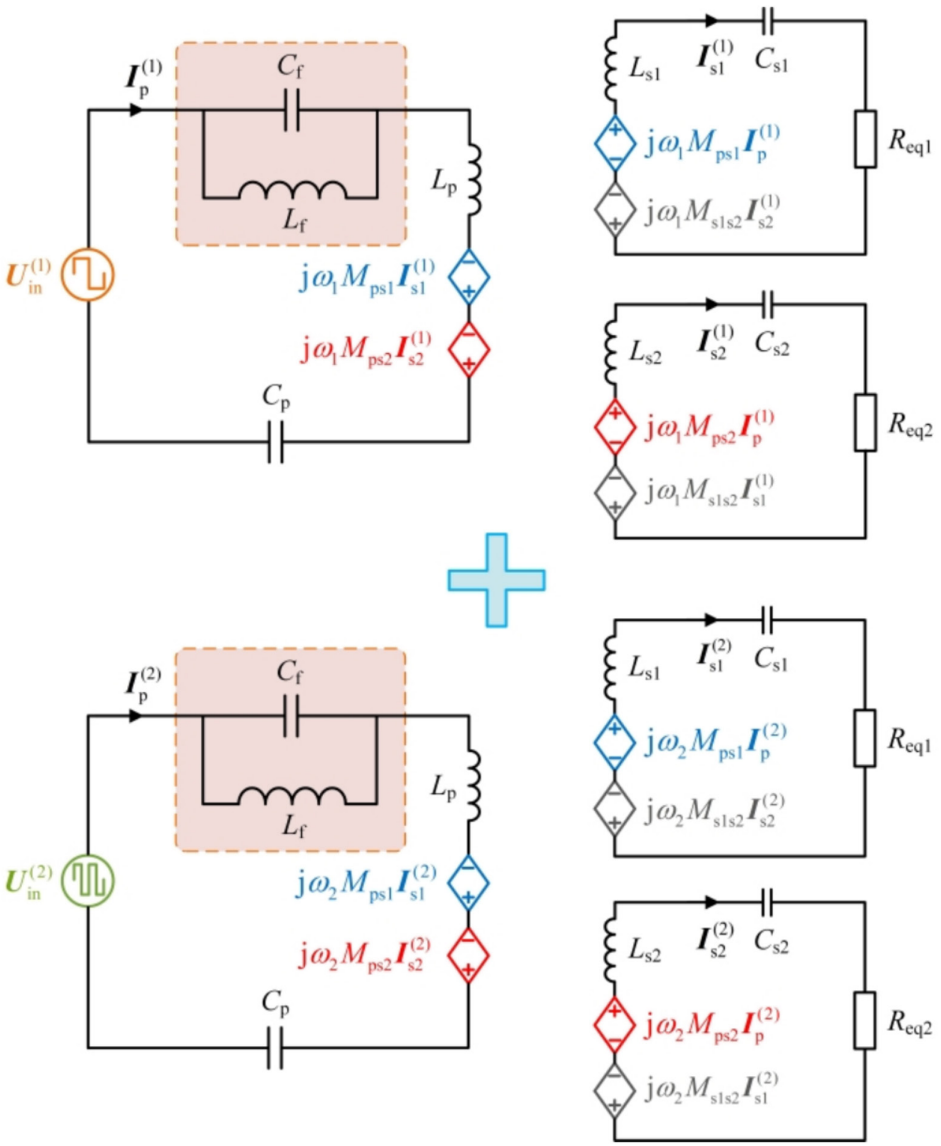


FIGURE 3 | Superposition-based equivalent circuit of the dual-frequency resonance OWPT system.

RX_{s2} are R_p, R_{s1}, and R_{s2}, respectively, the power loss of the system at different frequencies can be expressed as

$$\begin{cases} P_{loss}^{(1)} = R_p\left(I_p^{(1)}\right)^2 + \left(I_{s1}^{(1)}\right)^2 R_{s1} + \left(I_{s2}^{(1)}\right)^2 R_{s2} \\ P_{loss}^{(2)} = R_p\left(I_p^{(2)}\right)^2 + \left(I_{s1}^{(2)}\right)^2 R_{s1} + \left(I_{s2}^{(2)}\right)^2 R_{s2} \\ P_{loss} = P_{loss}^{(1)} + P_{loss}^{(2)} \end{cases} \tag{6}$$

where $P_{loss1}^{(1)}$ and $P_{loss1}^{(2)}$ represent the power loss of the transmitting and receiving circuits at frequencies f_1 and f_2, respectively, and P_{loss} is the total power loss of the transmitting and receiving circuits.

According to Equations (5) and (6), the system transmission efficiency can be expressed as

$$\eta = \frac{P_{o1} + P_{o2}}{P_{o1} + P_{o2} + P_{loss}} \tag{7}$$

2.2 | Topology Parameter Design

According to the transmission characteristics of the SS compensation topology, to reduce the reactive power loss of the system, the system's transmitting circuit is in a fully resonant state at both operating frequencies f_1 and f_2, while the receiving circuits are resonant at the respective frequencies f_1 and f_2. To ensure that the receiving circuits are in a resonant state, the following conditions must be met:

$$\begin{cases} C_{s1} = \dfrac{1}{\omega_1^2 L_{s1}} \\ C_{s2} = \dfrac{1}{\omega_2^2 L_{s2}} \end{cases} \tag{8}$$

To ensure that the transmitting circuit is fully resonant at both frequencies, it follows from Equation (2) that

$$Z_p^{(i)} = j\frac{\omega_i^2\left(L_pC_p + L_fC_f + L_fC_p\right) - \omega_i^4 L_pL_fC_pC_f - 1}{\omega_i C_p\left(1 - \omega_i^2 L_fC_f\right)} \tag{9}$$

Because the transmitting circuit needs to achieve full resonance at both operating frequencies, to further determine the values of compensation capacitors C_p and C_f, their relationship with frequencies f_1 and f_2 must be established. According to Equation (9), it can be derived that

$$f_1 = \frac{1}{2\pi\sqrt{L_p C_p}}\sqrt{\frac{1}{2}+\frac{\beta}{2}\left(1+\alpha-\sqrt{\left(\frac{\alpha\beta-1}{\beta}\right)^2+2\left(\frac{\alpha\beta+1}{\beta}\right)+1}\right)} \tag{10}$$

$$f_2 = \frac{1}{2\pi\sqrt{L_f C_f}}\sqrt{\frac{1}{2}+\frac{\beta}{2}\left(1+\alpha+\sqrt{\left(\frac{\alpha\beta-1}{\beta}\right)^2+2\left(\frac{\alpha\beta+1}{\beta}\right)+1}\right)} \tag{11}$$

where the parameters α, β, and γ are

$$\alpha = \frac{L_p}{L_f}, \quad \beta = \frac{C_p}{C_f}, \quad \gamma = \frac{f_2}{f_1} \tag{12}$$

Furthermore, the relationship that the compensation capacitors C_p and C_f should satisfy can be obtained as

$$\beta = \frac{C_p}{C_f} = \frac{\left(\left(\gamma^4+1\right)\alpha-2\gamma^2\right)\pm\sqrt{\left(\gamma^4-1\right)^2\alpha^2-4\gamma^2\left(\gamma^4+2\gamma^2+1\right)\alpha}}{2\left(\gamma^2\alpha^2+2\gamma^2\alpha+\gamma^2\right)} \tag{13}$$

To ensure that Equation (13) has a solution, the following conditions must be satisfied:

$$\alpha \geq \frac{4\gamma^2\left(\gamma^4+2\gamma^2+1\right)}{\left(\gamma^4-1\right)^2} \tag{14}$$

In addition, to avoid the impact of small-value capacitors on system performance, the value of β should satisfy the following condition based on the calculation results of Equation (13):

$$\beta = \max\{\beta\} \tag{15}$$

Therefore, the values of compensation capacitors C_p and C_f can be obtained from Equations (10), (11), and (15).

2.3 | Analysis of Power Transmission Characteristics

For the proposed dual-frequency resonant system, the square wave voltages $\boldsymbol{U}_{in}^{(1)}$ and $\boldsymbol{U}_{in}^{(2)}$ output by the inverter are given by

$$\begin{cases} \boldsymbol{U}_{in}^{(1)} = U_{dc1}\,\mathrm{sign}\left(\sin\left(2\pi f_1 t\right)\right) \\ \boldsymbol{U}_{in}^{(2)} = U_{dc2}\,\mathrm{sign}\left(\sin\left(2\pi f_2 t\right)\right) \end{cases} \tag{16}$$

where U_{dc1} and U_{dc2} represent the DC power supplies corresponding to frequency $f_{(i)}$, and the sign function is defined as

$$\mathrm{sign} = \begin{cases} 1, & \dfrac{n}{2f_{(i)}} \leq t \leq \dfrac{n+1}{2f_{(i)}} \quad (n = \text{Even Number}) \\ -1, & \dfrac{n}{2f_{(i)}} \leq t \leq \dfrac{n+1}{2f_{(i)}} \quad (n = \text{Odd Number}) \end{cases} \tag{17}$$

From Equation (16), it can be seen that to meet the power transmission requirements of different CEP devices, the amplitude of the DC input voltage at the corresponding frequency can be adjusted, thereby changing the amplitude of the input power at that frequency, achieving power adjustment for different loads.

Based on Figure 3, the equivalent impedance model of the system can be further obtained, as shown in Figure 4. Under the excitation of the voltage $\boldsymbol{U}_{in}^{(1)}$ at the operating frequency f_1, the corresponding equivalent impedances include the transmitting circuit impedance $Z_p^{(1)}$, the reflected impedance of receiving circuit 1 mapped to the transmitting side $Z_{ref\text{-}s1}^{(1)}$, and the reflected impedance of receiving circuit 2 mapped to the transmitting side $Z_{ref\text{-}s2}^{(1)}$. Similarly, under the excitation of the voltage $\boldsymbol{U}_{in}^{(1)}$ at the operating frequency f_2, the three corresponding equivalent impedances are $Z_p^{(2)}$, $Z_{ref\text{-}s1}^{(2)}$, and $Z_{ref\text{-}s2}^{(2)}$.

According to Equation (3), the expressions for $Z_{ref\text{-}s1}^{(i)}$ and $Z_{ref\text{-}s2}^{(i)}$ can be obtained as

$$\begin{cases} Z_{ref\text{-}s1}^{(i)} = \dfrac{\omega^2 M_{ps1}^2\left(R_{eq2}+Z_{s2}^{(i)}\right)-j\omega_i^3 M_{ps1}M_{ps2}M_{s1s2}}{\omega_i^2 M_{s1s2}^2+\left(R_{eq1}+Z_{s1}^{(i)}\right)\left(R_{eq1}+Z_{s2}^{(i)}\right)} \\ Z_{ref\text{-}s2}^{(i)} = \dfrac{\omega_i^2 M_{ps2}^2\left(R_{eq1}+Z_{s1}^{(i)}\right)-j\omega_i^3 M_{ps1}M_{ps2}M_{s1s2}}{\omega_i^2 M_{s1s2}^2+\left(R_{eq1}+Z_{s1}^{(i)}\right)\left(R_{eq1}+Z_{s2}^{(i)}\right)} \end{cases} \tag{18}$$

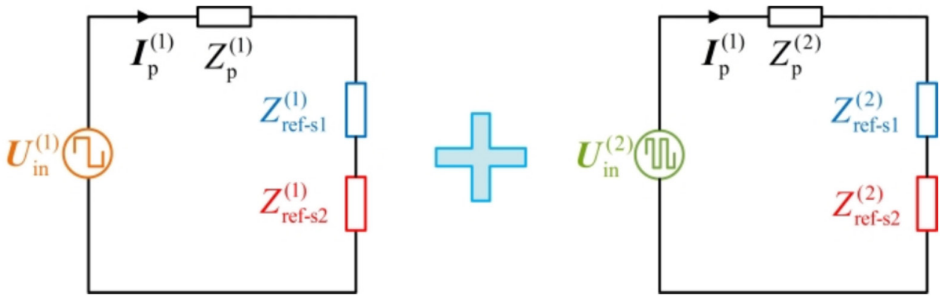


FIGURE 4 | Equivalent impedance model of the dual-frequency resonance OWPT system.

Furthermore, the ratio of the magnitudes of the reflected impedances of the two receiving circuits can be obtained as

$$\frac{\left|Z_{\text{ref-s1}}^{(i)}\right|}{\left|Z_{\text{ref-s2}}^{(i)}\right|}=\frac{\left|\omega_i^2M_{\text{ps1}}^2\left(R_{\text{eq2}}+Z_{\text{s2}}^{(i)}\right)-\text{j}\omega_i^3M_{\text{ps1}}M_{\text{ps2}}M_{\text{s1s2}}\right|}{\left|\omega_i^2M_{\text{ps2}}^2\left(R_{\text{eq1}}+Z_{\text{s1}}^{(i)}\right)-\text{j}\omega_i^3M_{\text{ps1}}M_{\text{ps2}}M_{\text{s1s2}}\right|} \tag{19}$$

TABLE 1 | Simulation model parameters.

Parameter	Description	Value
$U_{\text{in1}}/U_{\text{in2}}$	Input voltage	30 V
f_1/f_2	Switching frequency	40 kHz/100 kHz
L_{p}	TX inductance	48 μH
L_{f}	Primary compensation inductance	24 μH
L_{s1}	TXs1 inductance	102 μH
L_{s2}	TXs2 inductance	103 μH
C_{p}	Primary side series compensation capacitor	195.80 nF
C_{f}	Primary side parallel compensation capacitor	191.60 nF
C_{s1}	Compensation capacitance of receiving circuit 1	155.21 nF
C_{s2}	Compensation capacitance of receiving circuit 2	24.59 nF
R_{eq1}	Load resistance 1	2 Ω
R_{eq2}	Load resistance 2	2 Ω

Assume that the parameters of RX_{s1} and RX_{s2} are identical and their relative distances to the TX are equal. Therefore, Equation (19) can be simplified to

$$\frac{\left|Z_{\text{ref-s1}}^{(i)}\right|}{\left|Z_{\text{ref-s2}}^{(i)}\right|}=\frac{\left|\left(R_{\text{eq2}}+Z_{\text{s2}}^{(i)}\right)-\text{j}\omega_iM_{\text{s1s2}}\right|}{\left|\left(R_{\text{eq1}}+Z_{\text{s1}}^{(i)}\right)-\text{j}\omega_iM_{\text{s1s2}}\right|} \tag{20}$$

For the excitation component at the system's operating frequency f_1, $Z_{\text{s1}}^{(1)}=0$, indicating that receiving circuit 1 is in a fully resonant state; however, $Z_{\text{s2}}^{(1)}$ is in a nonresonant state, and its magnitude is relatively large. Therefore, $Z_{\text{ref-s1}}^{(1)}$ is much greater than $Z_{\text{s2}}^{(1)}$, meaning that the power generated by voltage $\boldsymbol{U}_{\text{in}}^{(1)}$ is mainly transmitted to the load R_{eq1}. Similarly, the power generated by voltage $\boldsymbol{U}_{\text{in}}^{(2)}$ is mainly transmitted to the load R_{eq2}. In summary, the system achieves dual-frequency, dual-channel energy transmission, and by adjusting the amplitudes of voltages $\boldsymbol{U}_{\text{in}}^{(1)}$ and $\boldsymbol{U}_{\text{in}}^{(2)}$, it can flexibly meet the transmission requirements of different power and frequency needs of the CEP equipment.

3 | Model Analysis of the Dual-Frequency Power Allocation Method

Based on the above analysis results, a simulation model of the dual-frequency resonant OWPT system was built using simulation software to verify the correctness and effectiveness of the proposed power allocation method. The key simulation parameters are shown in Table 1.

The dual-frequency resonant OWPT system proposed in this paper can be equivalently modeled as a dual-channel band-pass filter circuit. For example, at 40 and 100 kHz, the input voltage exhibits band-pass characteristics, while it shows high impedance characteristics at other frequencies. Figure 5 shows the simulation waveforms of the inverter output voltage $\boldsymbol{U}_{\text{in}}$ and output current $\boldsymbol{I}_{\text{in}}$ of the dual-frequency resonant OWPT system, with $\boldsymbol{U}_{\text{in}}^{(1)}=\boldsymbol{U}_{\text{in}}^{(2)}=30$ V.

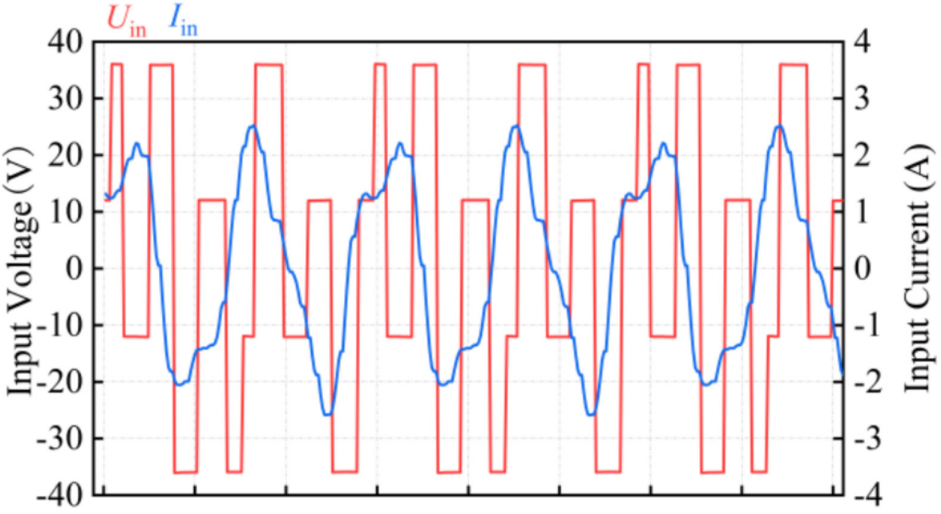


FIGURE 5 | Simulation waveforms of the input voltage and current.

Performing a fast Fourier transform (FFT) spectrum analysis on the results shown in Figure 5, the results are shown in Figure 6a. To demonstrate that the amplitudes of voltage excitations at two different frequencies can be independently adjusted, a control group simulation was added. The control groups were as follows: $\boldsymbol{U}_{\text{in}}^{(1)} = 40\,\text{V}$, $\boldsymbol{U}_{\text{in}}^{(2)} = 30\,\text{V}$; $\boldsymbol{U}_{\text{in}}^{(1)} = 50\,\text{V}$, $\boldsymbol{U}_{\text{in}}^{(2)} = 30\,\text{V}$; $\boldsymbol{U}_{\text{in}}^{(1)} = 30\,\text{V}$, $\boldsymbol{U}_{\text{in}}^{(2)} = 40\,\text{V}$. The results of the spectrum analysis are shown in Figure 6b–d. From the results, it can be seen that the coil input voltage and current mainly consist of 40- and 100-kHz components. Therefore, by changing the amplitude of the voltage excitation at different frequencies, the corresponding input voltage and current change

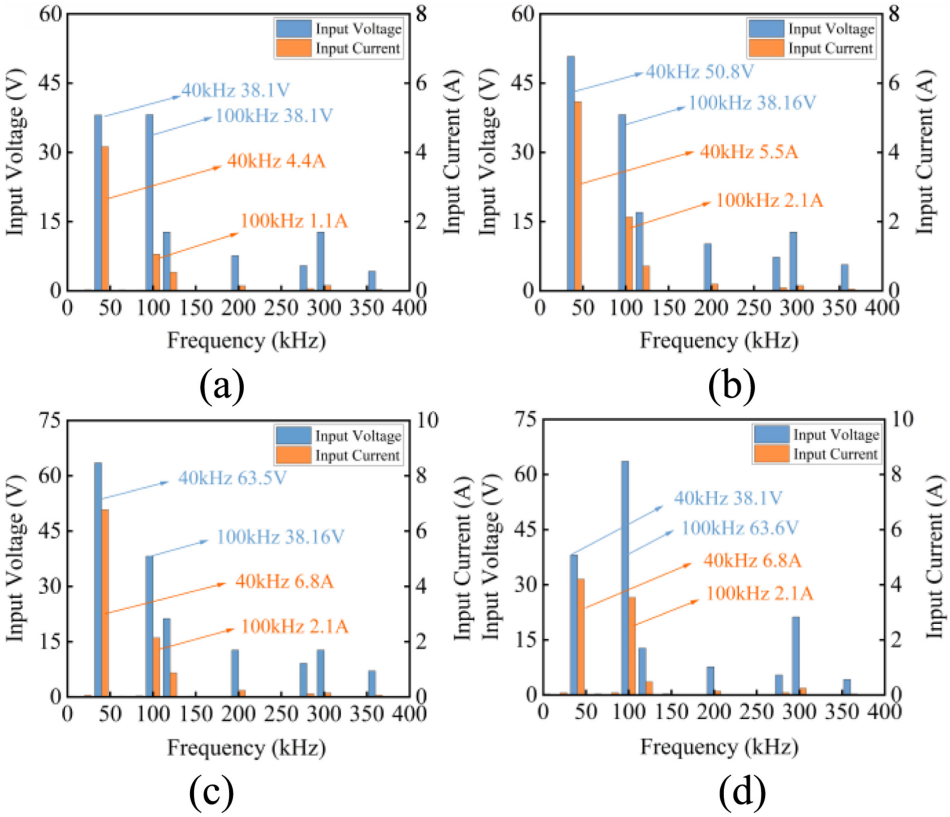


FIGURE 6 | FFT simulation results of input voltage and current. (a) $\boldsymbol{U}_{\text{in}}^{(1)} = \boldsymbol{U}_{\text{in}}^{(2)} = 30\,\text{V}$. (b) $\boldsymbol{U}_{\text{in}}^{(1)} = 40\,\text{V}$, $\boldsymbol{U}_{\text{in}}^{(2)} = 30\,\text{V}$. (c) $\boldsymbol{U}_{\text{in}}^{(1)} = 50\,\text{V}$, $\boldsymbol{U}_{\text{in}}^{(2)} = 30\,\text{V}$. (d) $\boldsymbol{U}_{\text{in}}^{(1)} = 30\,\text{V}$, $\boldsymbol{U}_{\text{in}}^{(2)} = 50\,\text{V}$.

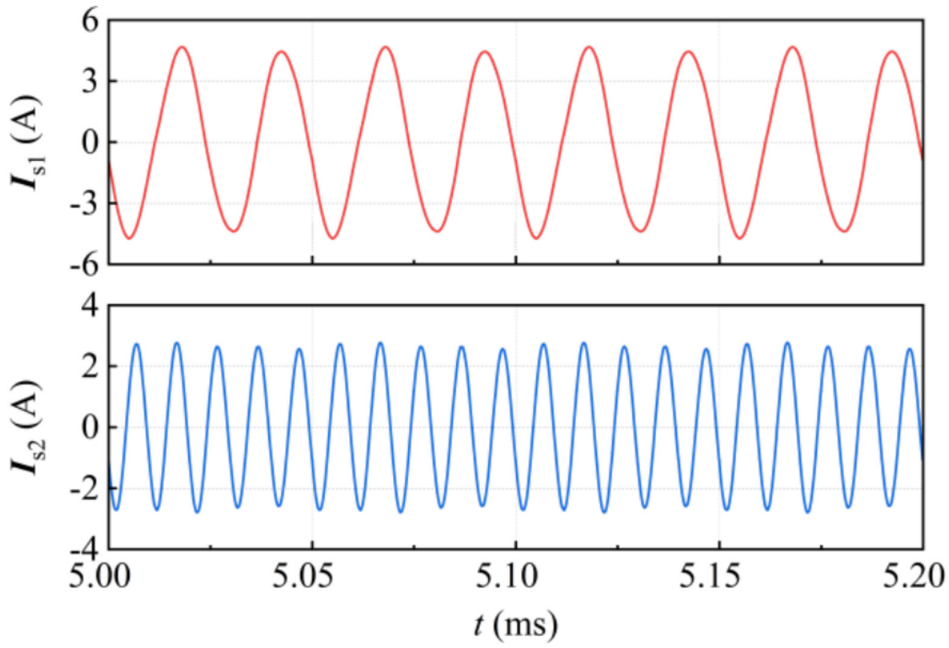


FIGURE 7 | Simulation waveforms of two receiver loop currents.

accordingly, allowing independent adjustment of the input power at the corresponding frequencies. From the results, it can be seen that the coil input voltage and current mainly consist of 40- and 100-kHz components. Therefore, by changing the amplitude of the voltage excitation at different frequencies, the corresponding input voltage and current change accordingly, allowing independent adjustment of the input power at the corresponding frequencies.

The simulation waveforms of the currents in the two receiving circuits are shown in Figure 7. Here, $\boldsymbol{I}_{s1}$ and $\boldsymbol{I}_{s2}$ represent the current waveforms of the receiving circuits with resonant frequencies of 40 and 100 kHz, respectively. The simulation results indicate that the output currents of the two receiving circuits with different frequencies are both sinusoidal, with smooth waveforms and relatively stable amplitudes.

In the dual-frequency OWPT system, with the input voltage kept constant, when the load resistance R_{eq2} remains constant and the load resistance R_{eq1} changes, the simulated current waveforms of the two receiving circuits are shown in Figure 8. Here, $\boldsymbol{I}_{s1}$ and $\boldsymbol{I}_{s2}$ represent the current waveforms of the receiving circuits with resonant frequencies of 40 and 100 kHz, respectively. When the load resistance R_{eq1} increases from 2 to 5 Ω, the output current

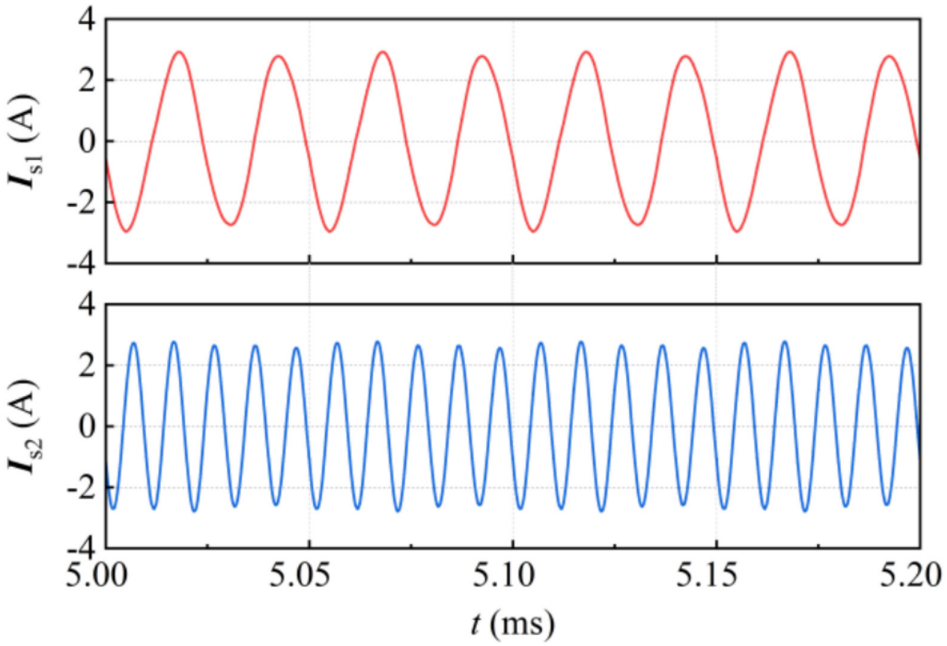


FIGURE 8 | The waveform of the two receiving currents varies with the load.

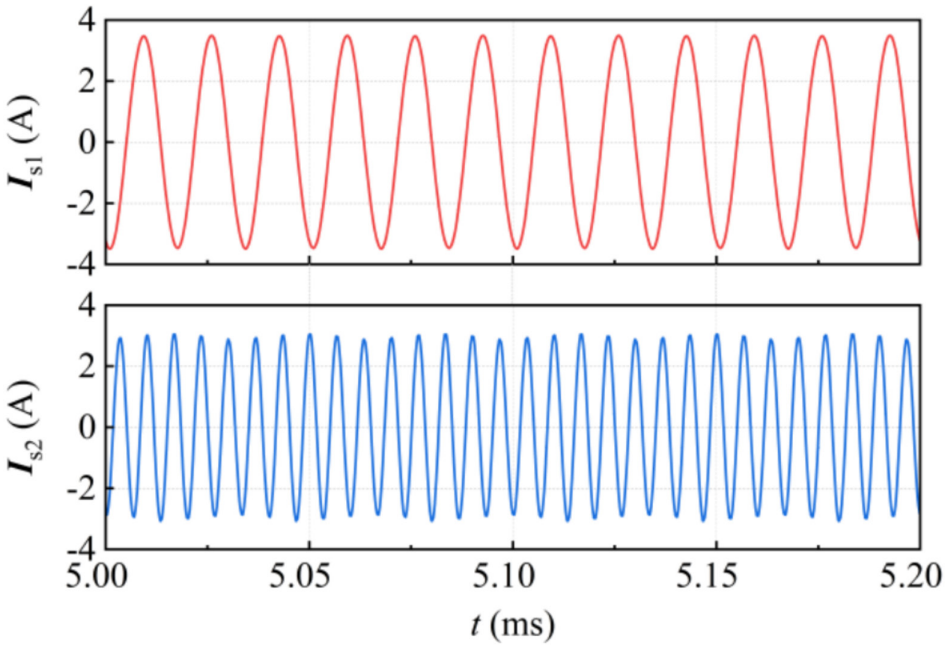


FIGURE 9 | The waveform of two receiving currents varies with frequency.

of receiving circuit 1 changes accordingly, decreasing from approximately 4.5 to 3 A, while the output current of receiving circuit 2 remains unchanged. The simulation results indicate that the two receiving circuits of the system are independent of each other. When the load of one receiving circuit changes, the output current of that receiving circuit changes accordingly, while the output current of the other receiving circuit is not affected.

To verify the applicability of the system's dual-frequency characteristics, the voltage excitation frequencies were set to 60 and 150 kHz, while simultaneously changing the natural resonant frequencies of the two receiving circuits to 60 and 150 kHz. With the system's input voltage kept constant, the simulated current waveforms of the two receiving circuits are shown in Figure 9. Here, $\boldsymbol{I}_{s1}$ and $\boldsymbol{I}_{s2}$ represent the current waveforms of the receiving circuits with resonant frequencies of 60 and 150 kHz, respectively. The simulation results indicate that the output currents of the two receiving circuits at different frequencies are both sinusoidal, with smooth waveforms and relatively stable amplitudes. Therefore, this system is applicable to different frequencies, and the design method is universal.

The received power of each receiving circuit is shown in Figure 10. The results indicate that receiving circuits with different resonant frequencies can effectively receive the excitation energy at their respective frequencies. Specifically, the power generated by the AC voltage $\boldsymbol{U}_{in}^{(1)}$ is mainly transmitted to the receiving circuit with a resonant frequency of 40 kHz, while the power generated by the AC voltage $\boldsymbol{U}_{in}^{(2)}$ is mainly transmitted to the receiving circuit with a resonant frequency of 100 kHz. This verifies the correctness and effectiveness of the proposed method.

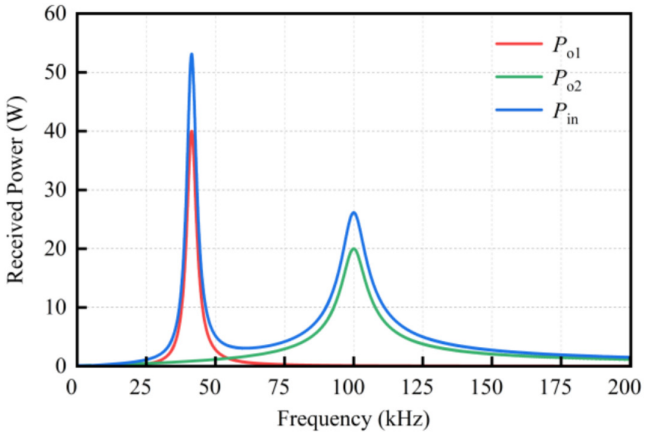


FIGURE 10 | System input and output powers.

4 | Experimental Validation

4.1 | System Setup

Figure 11 shows the experimental prototype of the OWPT system based on the dual-frequency resonance power allocation method. The transmitting side consists of a DC power supply,

TABLE 2 | Experimental prototype parameters.

Parameter	Description	Value
U_{in1}/U_{in2}	Input voltage	30 V
f_1/f_2	Switching frequency	39/98.5 kHz
L_p	TX inductance	48.5 μH
L_f	Primary compensation inductance	23.5 μH
L_{s1}	TXs1 inductance	103.2 μH
L_{s2}	TXs2 inductance	102.5 μH
C_p	Primary side series compensation capacitor	193.64 nF
C_f	Primary side parallel compensation capacitor	386.35 nF
C_{s1}	Compensation capacitance of receiving circuit 1	157.30 nF
C_{s2}	Compensation capacitance of receiving circuit 2	25.2 nF
R_{eq1}	Load resistance 1	2 Ω
R_{eq2}	Load resistance 2	2 Ω

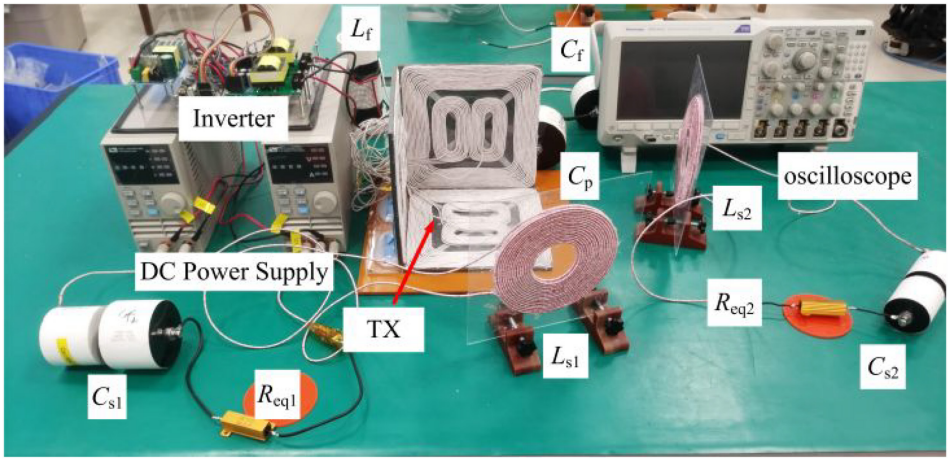


FIGURE 11 | Experimental prototype of the dual-frequency resonance OWPT system.

a full-bridge inverter circuit, and a TX composed of three orthogonal coils. The receiving side is composed of two sets of RX_{s1} and RX_{s2} with different natural resonances, compensation capacitors, and loads. The TX is made up of three orthogonal planar coils with magnetic cores and tight winding, connected in parallel to form an equivalent single coil. The RX are circular and tightly wound, with both the TX diameter and the RX outer diameter being 200 mm. The specific experimental parameters are shown in Table 2.

4.2 | Frequency Selection and Power Allocation

Based on the system parameters listed in Table 2, with both DC power supplies set to an amplitude of 30 V, the steady-state waveforms of the combined voltage $\boldsymbol{U}_{in}$ output by the inverter and the transmitting circuit current $\boldsymbol{I}_{p}$ under stable operating conditions are shown in Figure 12. The results are basically consistent with the simulation waveforms shown in Figure 5, indicating that the system input side can achieve dual-frequency composite excitation.

Figure 13 shows the dynamic variation waveforms of the receiver circuit currents $\boldsymbol{I}_{s1}$ and $\boldsymbol{I}_{s2}$ at two different resonant frequencies under various input voltage conditions. The testing process includes four stages: I: $\boldsymbol{U}_{in}^{(1)} = 30\,V$, $\boldsymbol{U}_{in}^{(2)} = 30\,V$; II: $\boldsymbol{U}_{in}^{(1)} = 40\,V$, $\boldsymbol{U}_{in}^{(2)} = 30\,V$; III: $\boldsymbol{U}_{in}^{(1)} = 50\,V$, $\boldsymbol{U}_{in}^{(2)} = 30\,V$; IV: $\boldsymbol{U}_{in}^{(1)} = 50\,V$, $\boldsymbol{U}_{in}^{(2)} = 40\,V$. The steady-state waveforms of $\boldsymbol{I}_{s1}$ and $\boldsymbol{I}_{s2}$ in each stage are shown in Figure 14a–d. The experimental results indicate that the excitation voltages at two different power levels achieved good superposition and transmitted power to the RX at the corresponding frequencies. Specifically, the resonant frequency of $\boldsymbol{I}_{s1}$ is 39 kHz, and the resonant frequency of $\boldsymbol{I}_{s2}$ is 98.5 kHz, confirming the receiver circuit's selectivity for the transmission power frequency.

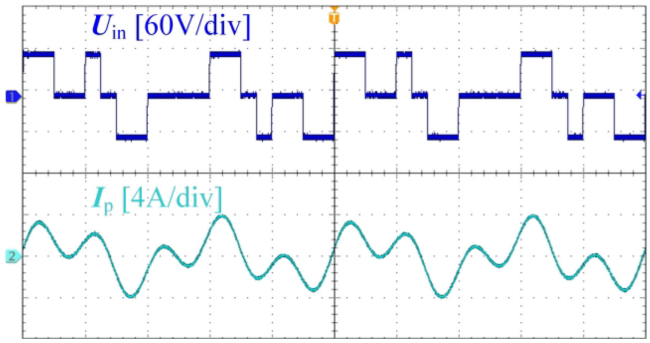


FIGURE 12 | Input voltage and current experimental waveform.

To verify that the system can meet the power demands of different loads and achieve independent power control, the output power and efficiency variations of two circuits were tested by changing the DC input voltage corresponding to the respective frequencies, as shown in Figure 15 and Figure 16. Taking receiving circuit 2 as an example, the amplitude of $\boldsymbol{U}_{in}^{(1)}$ in was kept constant, while the amplitude of $\boldsymbol{U}_{in}^{(2)}$ in was gradually changed in steps of 5 V. The results shown in Figure 15 indicate that when $\boldsymbol{U}_{in}^{(2)}$ in changes, the output power of receiving circuit 1 stabilizes around 40.42 W. While the output power of receiving circuit 2 gradually increases with the increase of $\boldsymbol{U}_{in}^{(2)}$ in, when $\boldsymbol{U}_{in}^{(2)} = 45\,V$, $P_{o2} = 29.13\,W$. Furthermore, as shown in Figure 16, the power transmission average efficiency of receiving circuit 2 is 74.9%, with a fluctuation rate ρ of only 4.97% as the amplitude of $\boldsymbol{U}_{in}^{(2)}$ in changes. Therefore, by adjusting the amplitude of the input voltage excitation, the output power of the receiving circuit at the corresponding frequency can be altered without affecting the output power at other frequencies, achieving power distribution and independent adjustment. This validates the correctness and effectiveness of the proposed dual-frequency distribution method.

The movement trajectory of the RX is shown in Figure 17. Among them, Trajectory 1 represents the RX rotating with a diameter of 150 mm from being perpendicular to L_{p3} to being perpendicular to L_{p3}. Trajectory 2 represents the RX rotating with a diameter of 150 mm from directly facing L_{p1} to directly facing L_{p2}. Meanwhile, the rotation centers of both trajectory 1 and trajectory 2 are at the physical center of the transmitting mechanism. Trajectory 1 and Trajectory 2 cover the typical positions of the RX. When there is only a single load in

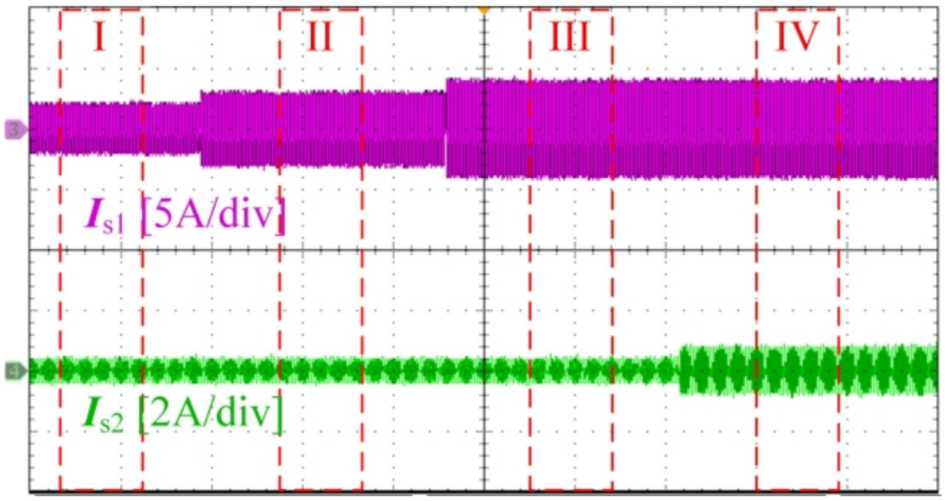


FIGURE 13 | Steady-state waveforms of output voltage and output current under dynamic conditions.

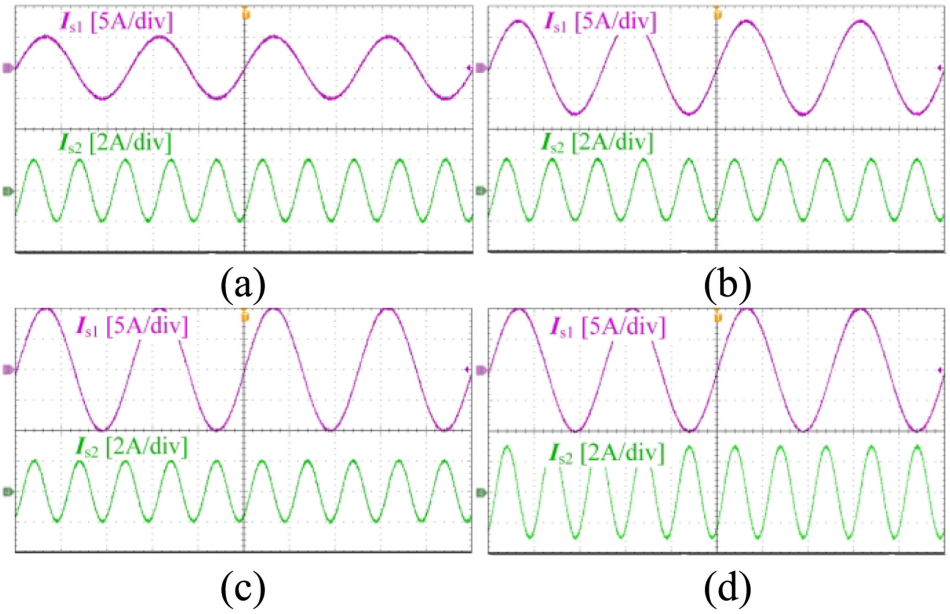


FIGURE 14 | Steady-state experimental waveforms of the output voltage and current: (a) Case I; (b) Case II; (c) Case III; (d) Case IV.

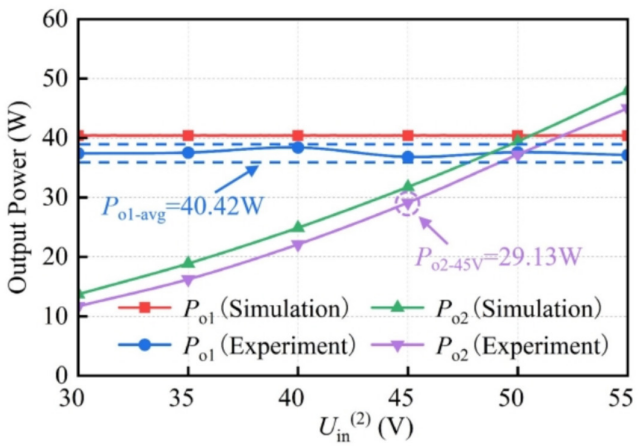


FIGURE 15 | Variations of the output power of the two receiver loops versus $\boldsymbol{U}_{in}^{(2)}$.

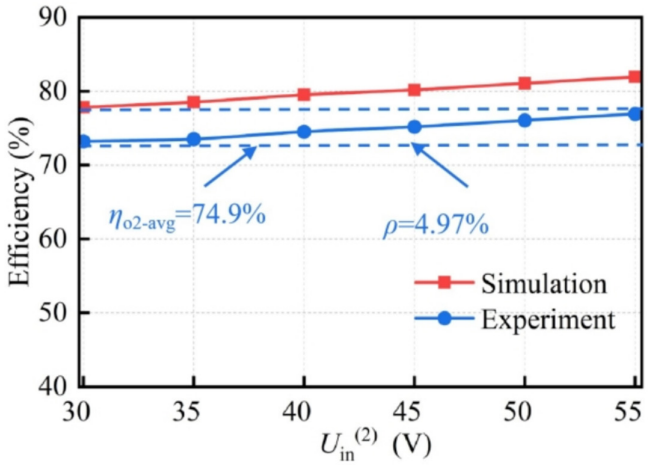


FIGURE 16 | Variations of system efficiency versus $\boldsymbol{U}_{in}^{(2)}$.

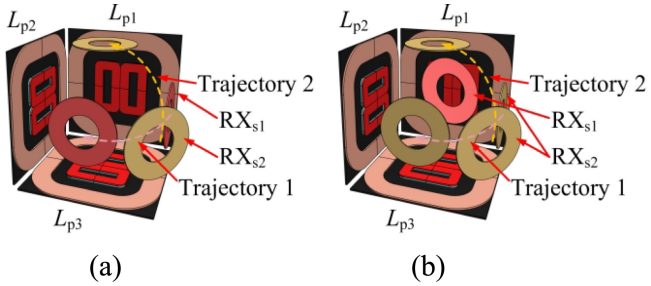


FIGURE 17 | Receiving coil movement trajectory. (a) Single receiving coil conditions. (b) Two receiving coil conditions.

the space, to verify that the RX can receive energy within the corresponding spatial region of the TX, the power transmission performance of the system was tested with the RX in different positions. RX_{s1} and RX_{s2} move along Trajectory 1 and Trajectory 2, respectively, as shown in Figure 17a. The output power and efficiency variations of the two receiving circuits are shown in Figure 18. The experimental results indicate that under the condition of a single receiving coil, the maximum output power of receiving circuit 1 is 38.85 W, with a minimum of 10.7 W, and the corresponding maximum and minimum efficiencies are 76.97% and 63.99%, respectively. The maximum and minimum output power of receiving circuit 2 are 15.42 and 8.7 W, with corresponding maximum and minimum efficiencies of 67.56%–72.57%, respectively. The experiments demonstrate that the RX can continuously and effectively capture energy in the space.

When there are two loads in the space, as shown in Figure 17b, the RX_{s1} and RX_{s2} are simultaneously placed in the working area. In the experiment, the RXs1 is in a fixed position, while the RXs2 moves around the TX along Trajectories 1 and 2. The system's total output power (the sum of P_{o1} and P_{o2}) and system efficiency are shown in Figure 19. The experimental results

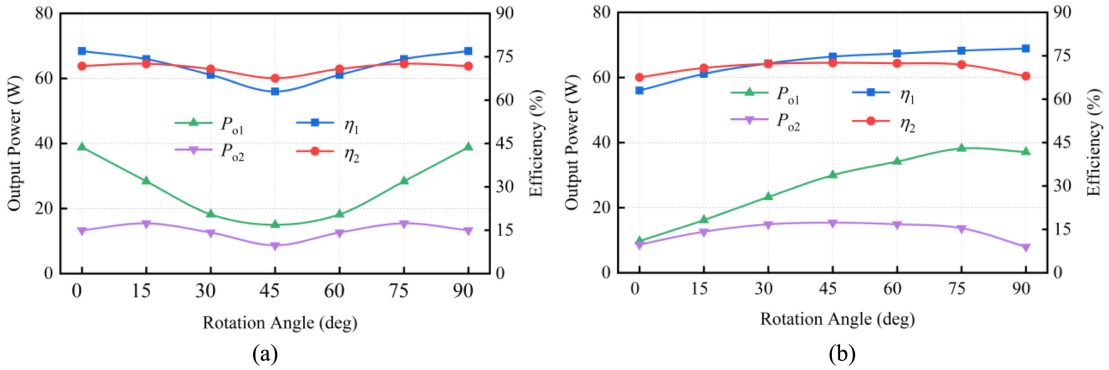


FIGURE 18 | Variations in output power and efficiency of a single receiving coil working along different trajectories. (a) Trajectory 1; (b) Trajectory 2.

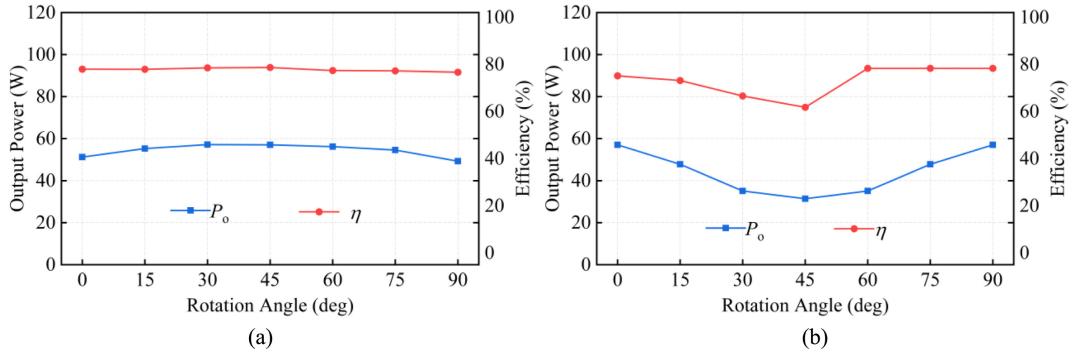


FIGURE 19 | Variations in output power and efficiency of two receiving coils working along different trajectories. (a) Trajectory 1. (b) Trajectory 2.

indicate that the system's maximum and minimum power are approximately 57.15 and 49.25 W, respectively, with corresponding maximum and minimum efficiencies of 78.68% and 76.67%. The proposed dual-frequency OWPT system can effectively transmit power to all target receiving coils within the power transfer space, verifying the correctness and feasibility of the proposed method.

5 | Conclusion

This paper proposes a novel dual-frequency power allocation method for OWPT systems suitable for CEP omnidirectional wireless charging systems. This method aims to achieve frequency division power supply and independent power regulation for CEP devices operating at different frequencies by designing an SS topology suitable for dual-frequency resonant circuits. Simulation results are presented and analyzed in detail, verifying the effectiveness of the designed method, which aligns well with theoretical analysis. A dual-frequency OWPT platform operating at 40 kHz/100 kHz was built to verify the correctness and effectiveness of the proposed power allocation method. Experimental results show that when the input voltage corresponding to only one frequency changes, only the output power of the load corresponding to that frequency changes accordingly. Meanwhile, under the condition of dual-load power supply, when the receiving coil is within a 90° offset range, the system's output power ranges from 49.25 to 57.15 W with a fluctuation rate of 14.8%, and the output efficiency ranges from 76.67% to 78.68% with a fluctuation rate of 2.59%.

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Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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